Chemical mediators and chronic inflammation

SGL4

Learning outcome

- Explain the chemical mediators and there effects on vascular and cellular changes during inflammation
- Define and classify chronic inflammation
- Recognize microscopic finding in granulomatous inflammation

Chemical mediator

- 1- They are substances that trigger both vascular & cellular reactions of inflammation
- 2- They initiate & regulate inflammatory reactions.
- 3- They determine clinical cardinal sign & pathologic manifestations of inflammation

General properties of chemical Mediators

- **1-Active mediators are produced only** in response to stimuli including microbial products & necrotic cells.
- **2-** The actions are **short lived & tightly regulated**
- **3-Most act by binding to specific receptors .**
- **Actions**:

❖Autocrine effect (act on same cell secreting mediator)

❖Paracrine (act on adjacent cell)

❖Endocrine effect through blood act distally

General properties of chemical Mediators

- **4- Some** have **direct enzymatic** or **toxic effect** that do not require specific receptor like **lysosomal enzyme & ROS**.
- **5- May have different & opposing activities-** amplifying or counteracting action of a mediator.
- **6- One mediator can stimulate release of other mediators** like products of complement activation stimulate the release of histamine & cytokine .

Chemical mediators origin

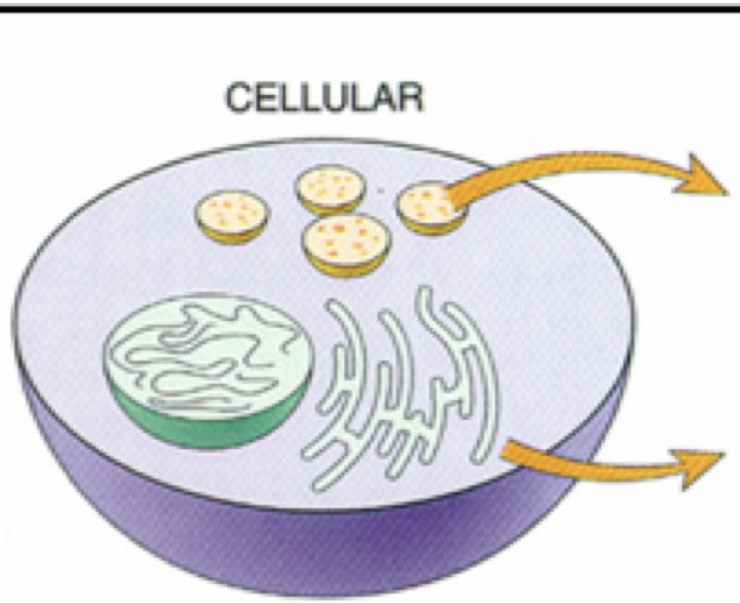
- **1-Cell derived :** locally produced by cell either
- Pre synthesized or Newly synthesized (ready made) .

The major cell types that produce mediators of acute inflammation are;

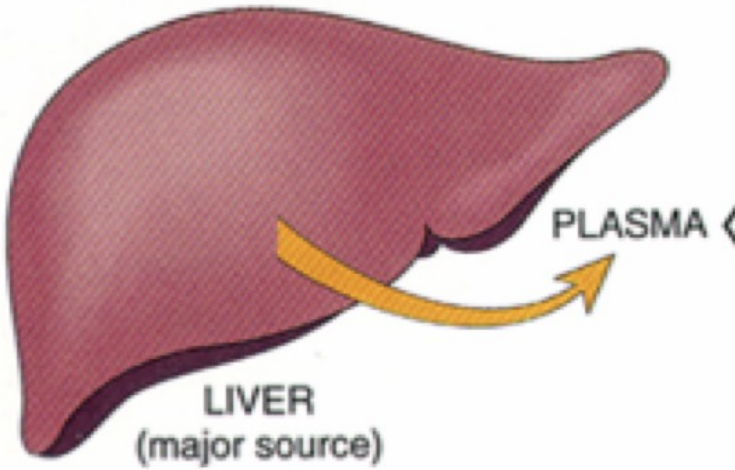
- **tissue macrophages & mast cells,** but platelet ,neutrophils & others also can elaborate some of mediators.

- 2- **Plasma derived** : from inactive precursors in plasma

Chemical mediators origin



CELLULAR



PLASMA

LIVER
(major source)

	MEDIATORS	SOURCE
Preformed mediators in secretory granules	Histamine	Mast cells, basophils, platelets
	Serotonin	Platelets
	Lysosomal enzymes	Neutrophils, macrophages
Newly synthesized	Prostaglandins	All leukocytes, platelets, EC
	Leukotrienes	All leukocytes
	Platelet-activating factors	All leukocytes, EC
	Activated oxygen species	All leukocytes
	Nitric oxide	Macrophages
	Cytokines	Lymphocytes, macrophages, EC
Factor XII (Hageman factor) activation	Kinin system (bradykinin)	
	Coagulation / fibrinolysis system	
Complement activation	C_{3a}, C_{5a}	anaphylatoxins
	C_{3b}	
	C_{5b-9}	(membrane attack complex)

1. Cell derived mediators

a) vasoactive amines (serotonin,histamine)

b) Arachidonic acid metabolites

- cycloxygenase pathway
- lipoxygenase pathway

c) Lysosomal components

d) Platelet activating factor

e)Cytokines (IL-1,TNF-α,TNF-β,IF-γ,Chemokines)

f) Nitric oxide and oxygen metabolites

a) Vasoactive amines

Histamine :- Histamine is an organic nitrogenous compound as well as regulating physiological function in the gut and acting as a neurotransmitter for the brain, spinal cord, and uterus.

- Mast cells are richest source of histamine, basophiles and platelets.
- Released by the stimuli of various agents like Heat, Cold, Irradiation, Irritant chemicals, Anaphilatoxins, Interleukins,.. etc.
- Actions ; ↑ Vasodilation
  Vascular permeability
  Itching and pain

Serotonin/5-hydroxy tryptamine :-

The scientific name for serotonin is 5-hydroxytryptamine, or 5-HT. It is mainly found in the chromafin cells of GIT, Spleen, Nervous tissue, Mast cells, Platelets. brain, bowels, and blood.

- Actions ; Similar to Histamine, but lesspotent
  - ⇧Vasodilation
  - Vascular permeability

b) Arachidonic acid Metabolites

Arachidonic acid is a polyunsaturated fatty **acid**

- produced by endothelial cells, leukocytes and platelets
- act locally on smooth muscle, endothelium and platelets
- origin:

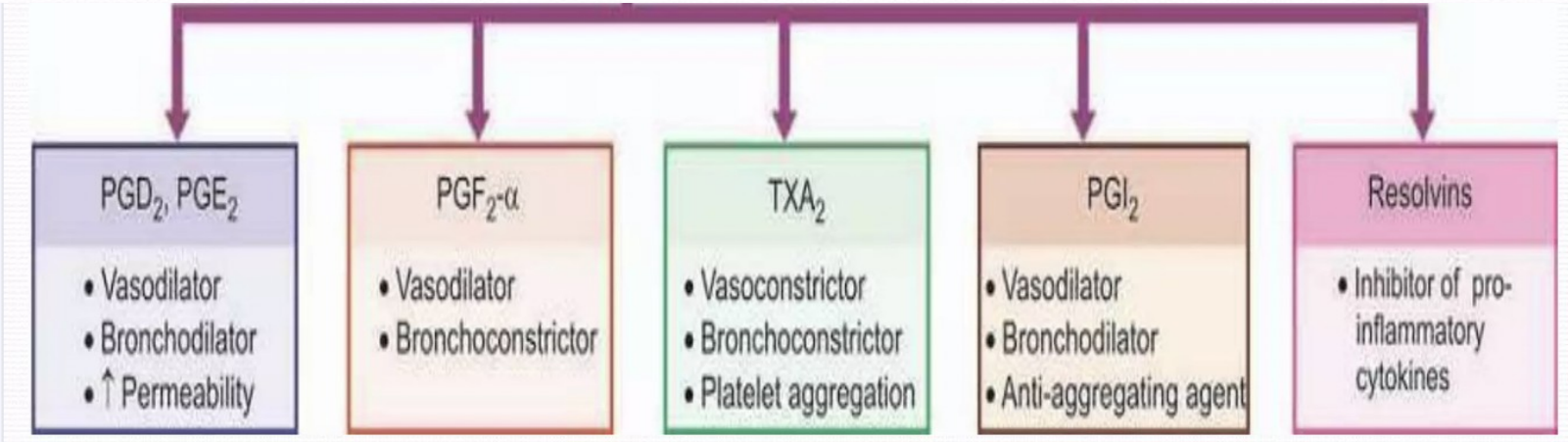
Arachidonic acid derived from dietary sources or by conversion

from the linoleic acid.

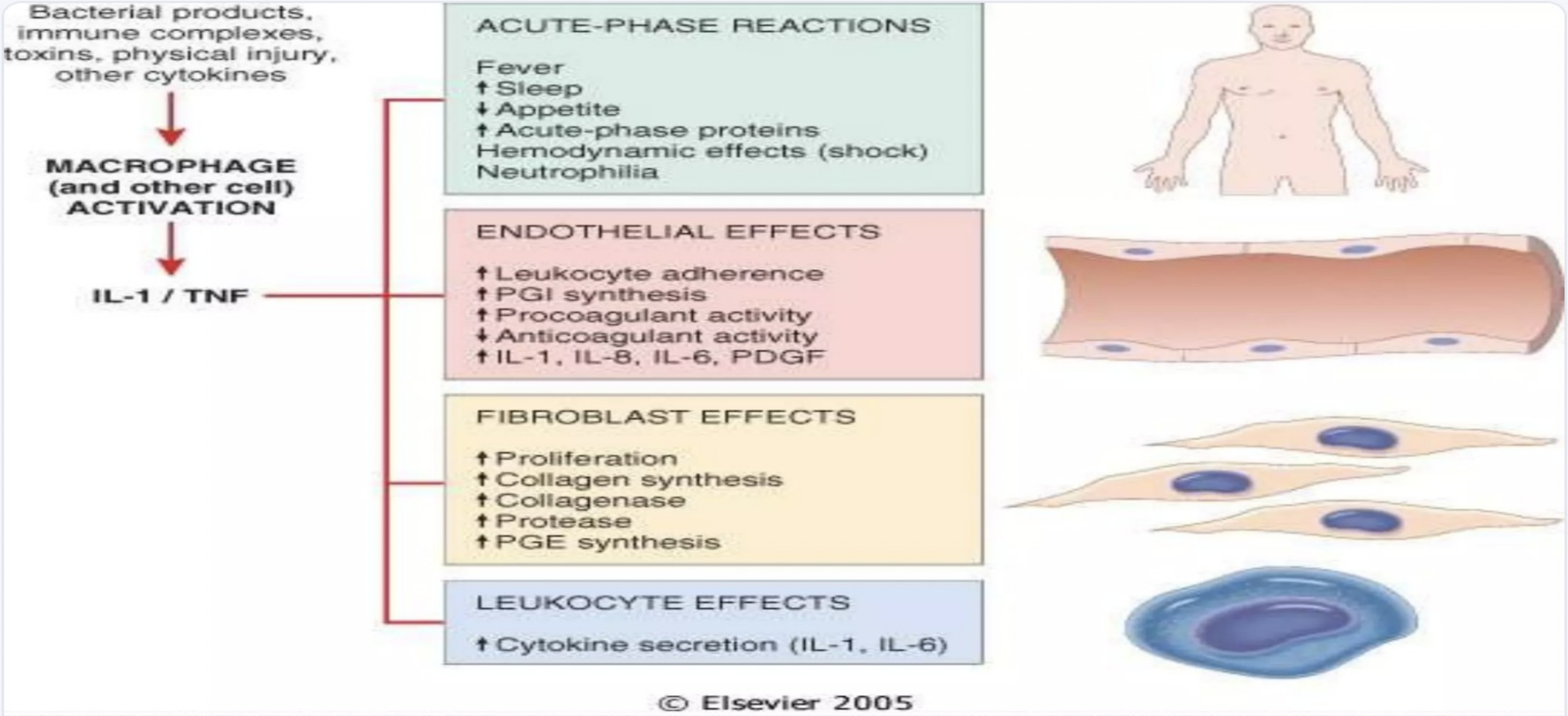
And released by activated phospholipases (in injury)

- two important pathways:
  - cyclooxygenase (COX)
  - lipoxygenase

- AA Metabolites
  - Cyclooxygenase pathway
    - PGD2,PGE2,PGF2α (prostaglandin)
      - V.D
      - Potentiates Edema formation
    - PGI2(prostacyclin)
      - Produced by prostacyclin synthase in endothelial cell
      - V.D, Inhibits Platelet aggregation
    - TxA2(throboxane A2)
      - Produced by Thromboxane synthase in pllatelets
      - stimulates platelets aggregation



- Lipoxygenase Pathway: present in neutrophils
  - LTB4
    - Produced by neutrophils & some macrophages
    - Chemotactic agent for neutrophils
  - LTC4,LTD4 & LTE4
    - Produced by mast cells
    - V.C bronchospasm
  - Lipoxins
    - Endogenous antagonists of Leukotrienes
  - PAF **Platelet-activating factor**, or AGEPC (acetyl-glyceryl-ether-phosphorylcholine), is a potent phospholipid **activator** and mediator of many leukocyte functions,**platelet** aggregation
    - Produced by WBCs & endothelial cells
    - Causes V.C, Bronchoconstriction



Major effects of interleukin-1 (IL-1) and tumor necrosis factor (TNF) in inflammation

CYTOKINES/CHEMOKINES

- **CYTOKINES**
  - POLYPEPTIDES produced by MANY cells (usually LYMPHOCYTES and MACROPHAGES)
  - Multiple, basic role in acute and chronic inflammation
  - **TNFα, IL-1** from macrophages
- **CHEMOKINES** are small polypeptides, attractants for PMNs (>40)

NITRIC OXIDE

- Potent vasodilator
- Produced from the action of nitric oxide synthetase from arginine

FREE RADICALS

- O_2 (SUPEROXIDE)
- H_2O_2 (PEROXIDE)
- OH^- (HYDROXYL RADICAL)
- **VERY-VERY DESTRUCTIVE**

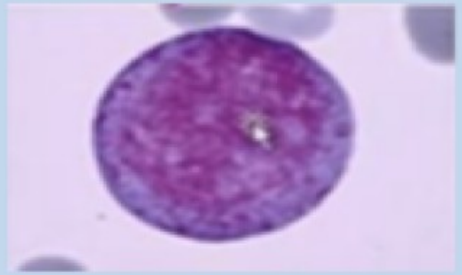
LYSOSOMAL COMPONENTS

PRIMARY

- = AZUROPHILIC, or NON-specific granules
- Myeloperoxidase (MPO)
- Lysozyme (bactericide)
- Acid hydrolases

SECONDARY

- SPECIFIC granules
- Lactoferrin
- Lysozyme
- Alkaline phosphatase
- Collagenase





2.Plasma derived mediators

a) The kinin system :

b) The clotting system :

c) The fibrinolytic system :

d) The complement system :

2: Plasma Protein derived mediators

a)Kinin System: The **kinin**– kallikrein **system** or simply **kinin system** is a poorly hormonal **system** .

It consists of blood proteins that play a role in inflammation, blood pressure control, coagulation and pain.

- Ultimately leads to formation of bradykinin
- Bradykinin causes
- Arteriolar dilation, increases vascular permeability & broncho constriction

Complement System: The complement system is a collection of soluble proteins

The **complement system** is a part of the immune **system**

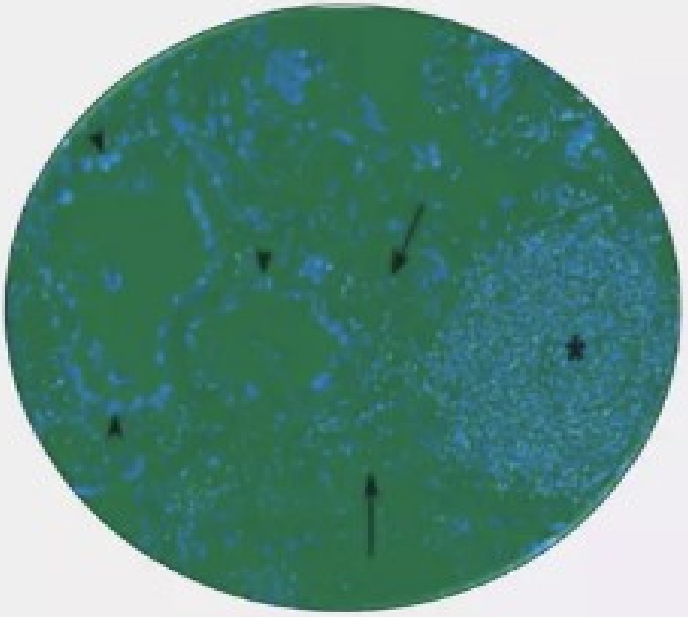
that enhances (**complements**) the ability of antibodies and phagocytic cells to microbes and damaged cells from an organism,

promotes inflammation, and attacks the pathogen's plasma membrane

- Mainly leads to the formation of C3a & C5a
- Vascular effects(C3a & C5a)
- Leukocyte activation, adhesion, chemotaxis(C5a)
- Phagocytosis(C3b,iC3b)

- Membrane attack complex cause pores in cell of invading microbes.

- Fate of acute inflammation?



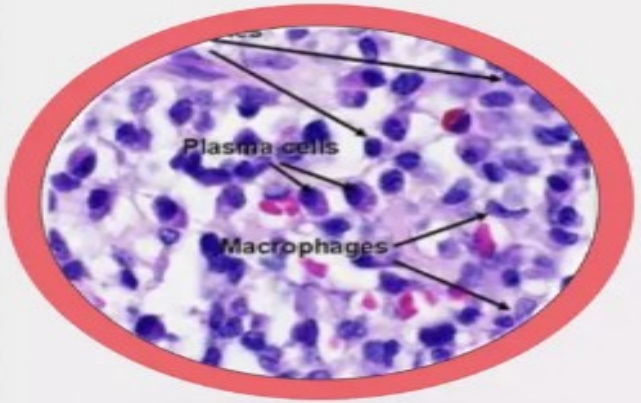
CHRONIC INFLAMMATION

Chronic inflammation is a response of prolonged duration (weeks or months) in which inflammation, tissue injury, and attempts at repair coexist, in varying combinations.

Chronic inflammation may follow acute inflammation If the inflammatory response fails to clear the cause,
or → may arise de novo .

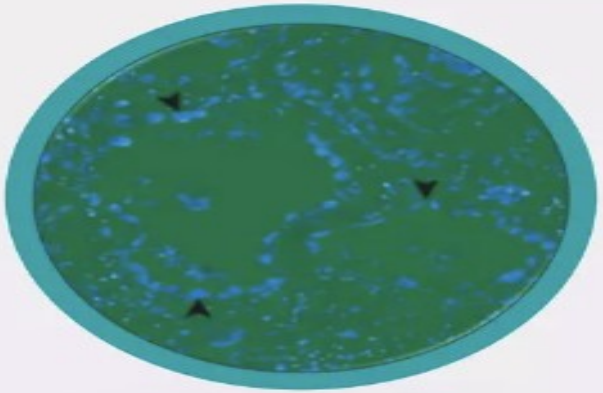


Essential features of chronic inflammation



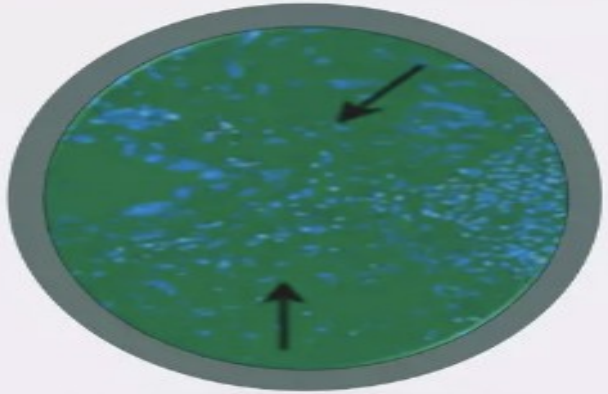
Infiltration with mononuclear cells

- Macrophages
- Lymphocytes and
- Plasma cells



Tissue Destruction

- Induced by persistent **offending agent i.e.,**
- Microorganism or
- **Inflammatory cells** like Macrophages



Attempts at healing

- By **connective tissue replacement** or deposition at the site of damaged tissue
- There is **Angiogenesis** and **fibrosis**



Dominant cells in Chronic Inflammatory reaction

Macrophages Distribution



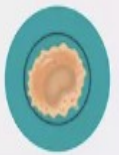
- Macrophages are tissue cells derived
  - From **hematopoietic stem cells** in the bone marrow and
  - From **progenitors** in the embryonic yolk sac and fetal liver during early development



- In circulation they are known as **monocytes.**
- Macrophages are normally diffusely scattered in most **connective tissues.**



- Found in specific locations in organs i.e,
  - Liver (**Kupffer** cells)
  - Spleen and lymph nodes (**sinus histiocytes**)
  - Central nervous system (**microglial cells**)
  - Lungs (**alveolar macrophages**)



- Together these cells comprise the **mononuclear phagocyte system**

❖ Function of macrophage

- 1-Phagocytosis
- 2-Initiate tissue repair
- 3- Secret inflammatory mediators : TNF,IL1,chemokine augment infl.
- 4- Anti-inflammatory .effect suppress inflammation.
- 5-Display Ag to T- lymphocytes

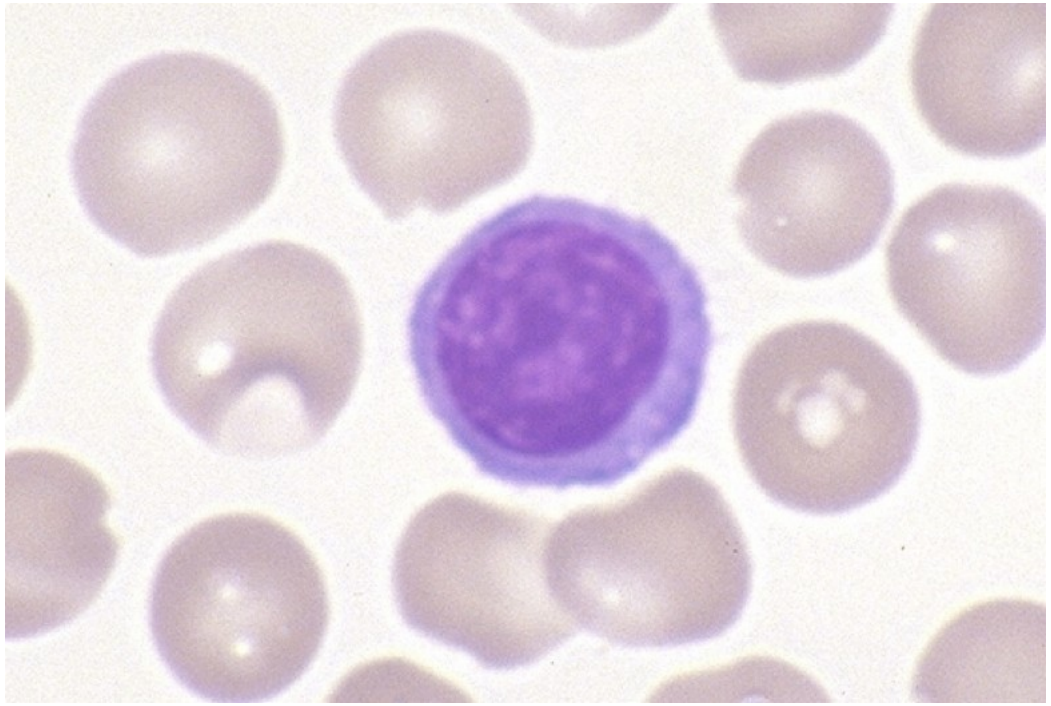
2. Role of Lymphocytes in Chronic Inflammation

- **Microbes** and other **environmental antigens** activate T and B lymphocytes, which amplify and propagate chronic inflammation

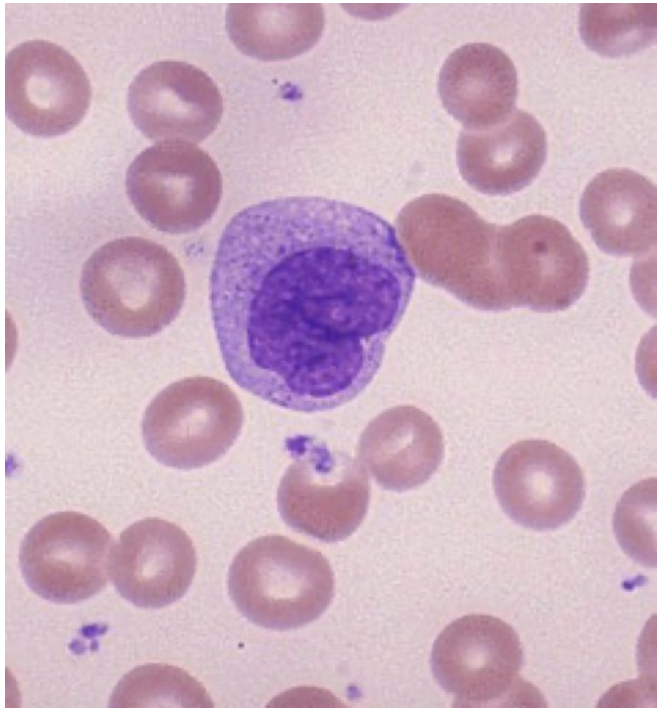
- Some of the strongest chronic inflammatory reactions, such as **granulomatous inflammation** are dependent on lymphocyte responses.

- Lymphocytes may be the dominant population in the chronic inflammation seen in **autoimmune** and other **hypersensitivity diseases**

Chronic inflammatory cells



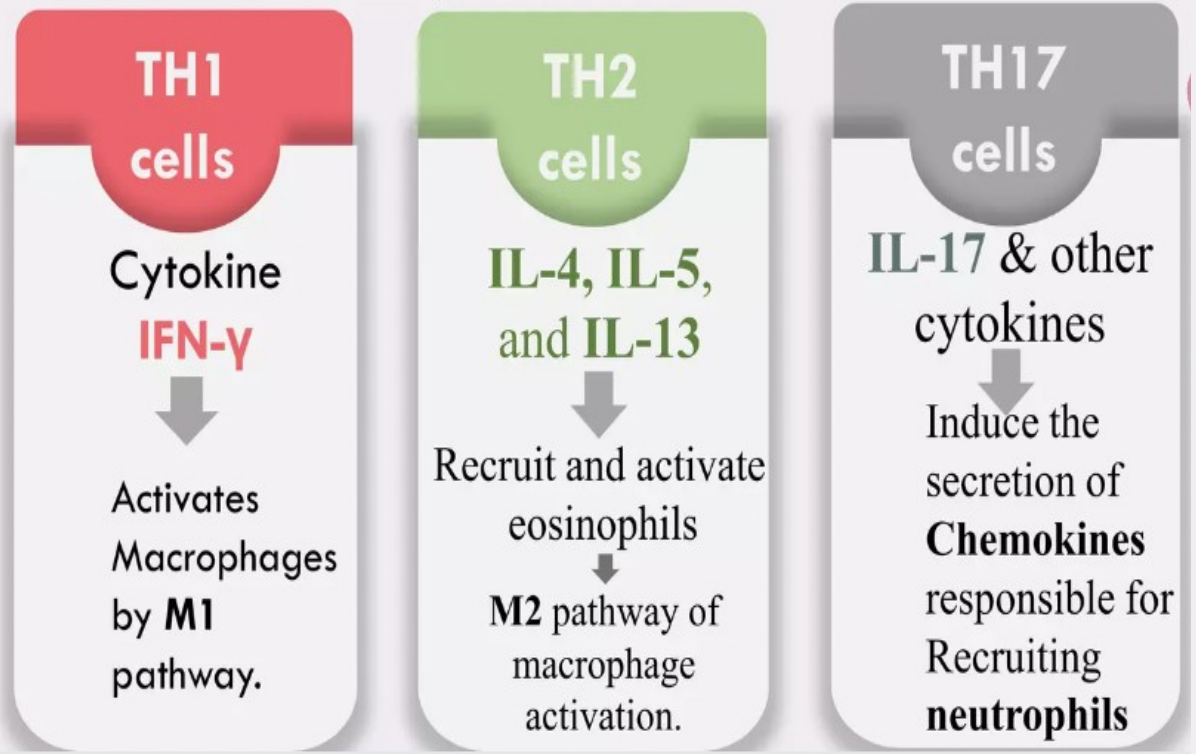
LYMPHOCYTE



“MONO”CYTE
MACROPHAGE
HISTIOCYTE

- **CD4+ T lymphocytes promote inflammation** by secreting cytokines, hence promote inflammation

- There are **three subsets** of CD4+ T cells that secrete different cytokines



- Both **TH1 and TH17** cells are involved in defense against many types of **bacteria** and **viruses** and in **autoimmune diseases.**

- **TH2 cells** are important in defense against **helminthic parasites** and in **allergic inflammation**

Lymphocytes and macrophages interact in a bidirectional way

Macrophages display

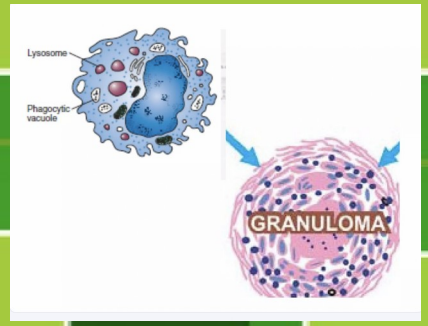
- Antigens to T cells
- Express membrane molecules (called costimulators) → activate T cells
- Produce cytokines (**IL-12** and others) that also stimulate T cell responses

Activated **T lymphocytes**

- Produce cytokines
- Recruit and activate macrophages
- Promoting more antigen presentation and cytokine secretion.
- The result is a cycle of cellular reactions that fuel and sustain chronic inflammation

Classification

Chronic inflammation



Chronic non specific inflammation

Chronic granulomatous inflammation

GRANULOMATOUS INFLAMMATION

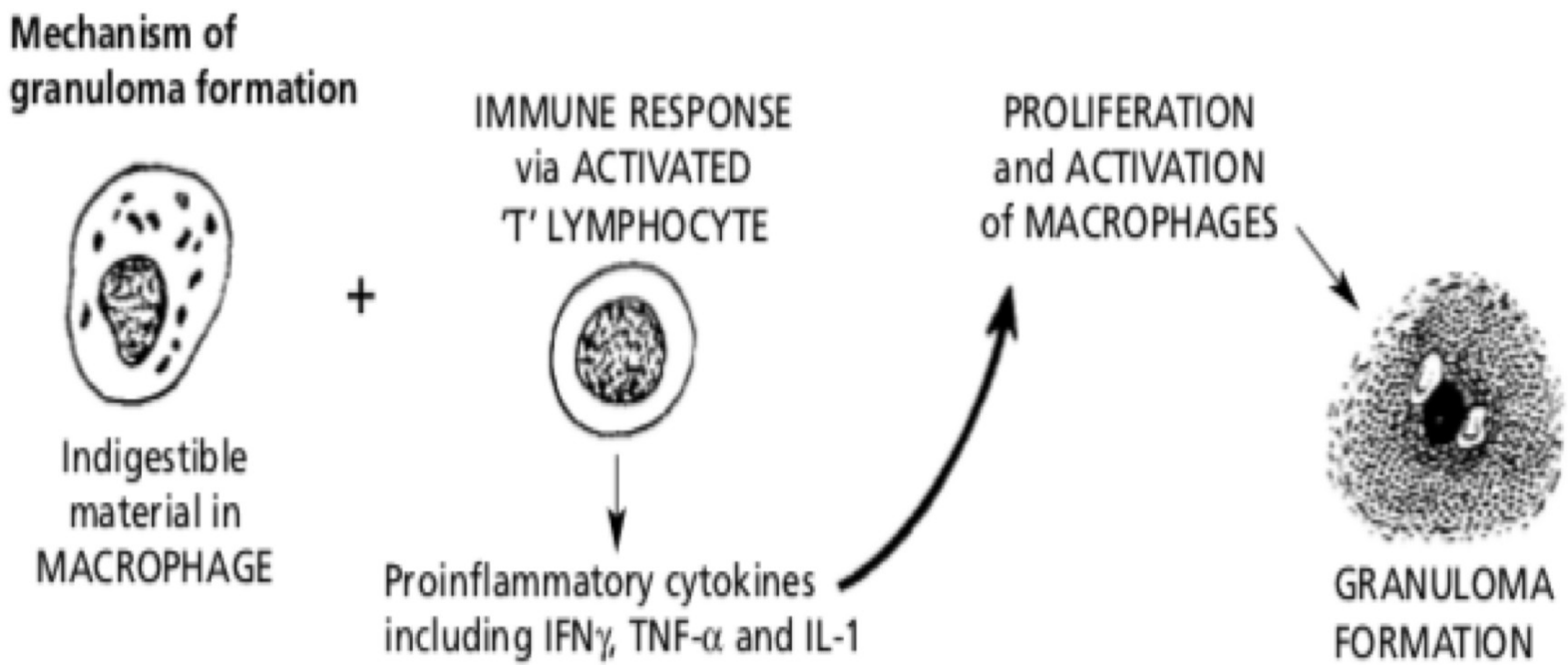
GRANULOMATOUS INFLAMMATION

***Granulomatous inflammation** is a distinctive pattern **of chronic inflammation** that is encountered in a limited number of infectious and some noninfectious conditions. **Immune reactions are usually involved in the development of granulomas**.*

A granuloma is a cellular attempt to control an offending agent that is difficult to eradicate.

.

- **In this attempt there is often strong activation of T lymphocytes leading to macrophage activation, which can cause injury to normal tissues**



Types of Granulomas

I. Foreign body granulomas

II. Immune granulomas

Foreign body granulomas

Incited by relatively inert foreign bodies.

Typically, foreign body granulomas form around material that are large enough to preclude phagocytosis by a single macrophage and do not incite any specific inflammatory or immune response.

Immune granulomas

Caused by agents that are capable of inducing an immune response which produces granulomas usually when the inciting agent is poorly degradable or particulate.

Example of disease with granulomatous inflammation

Disease	Cause	Tissue Reaction
Tuberculosis	*Mycobacterium tuberculosis*	Caseating granuloma (tubercle): focus of activated macrophages (epithelioid cells), rimmed by fibroblasts, lymphocytes, histiocytes, occasional Langhans giant cells; central necrosis with amorphous granular debris; acid-fast bacilli
Leprosy	*Mycobacterium leprae*	Acid-fast bacilli in macrophages; noncaseating granulomas
Syphilis	*Treponema pallidum*	Gumma: microscopic to grossly visible lesion, enclosing wall of histiocytes; plasma cell infiltrate; central cells necrotic without loss of cellular outline
Cat-scratch disease	Gram-negative bacillus	Rounded or stellate granuloma containing central granular debris and recognizable neutrophils; giant cells uncommon
Sarcoidosis	Unknown etiology	Noncaseating granulomas with abundant activated macrophages
Crohn disease (inflammatory bowel disease)	Immune reaction against intestinal bacteria, self-antigens	Occasional noncaseating granulomas in the wall of the intestine, with dense chronic inflammatory infiltrate

Giant Cells

Frequently, epithelioid cells fuse to form *giant cells* in the periphery or sometimes in the center of granulomas.

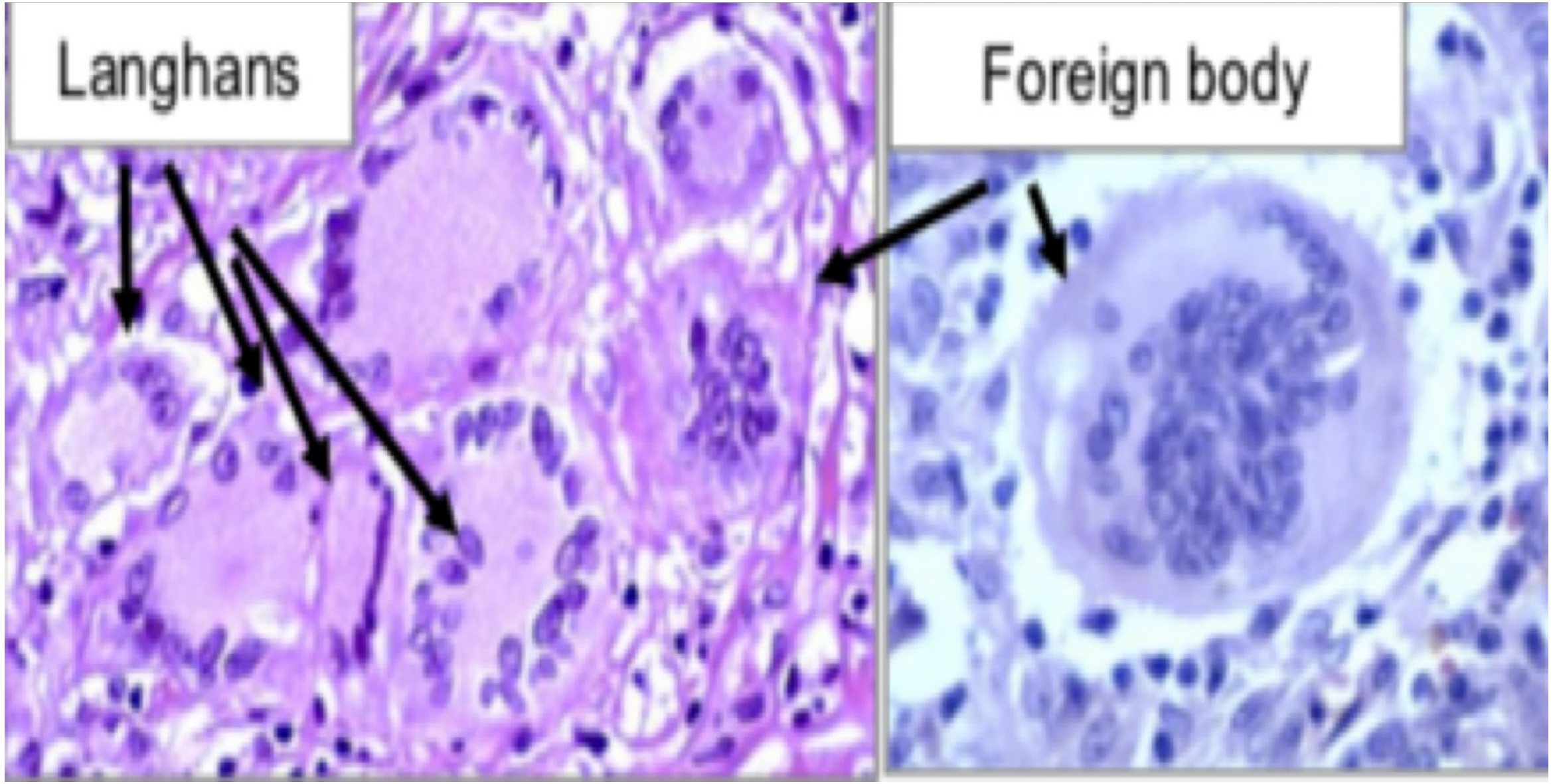
These giant cells may attain diameters of 40 to 50 µm.

peripherally (Langhans-type giant cell) or haphazardly (foreign body–type giant cell).

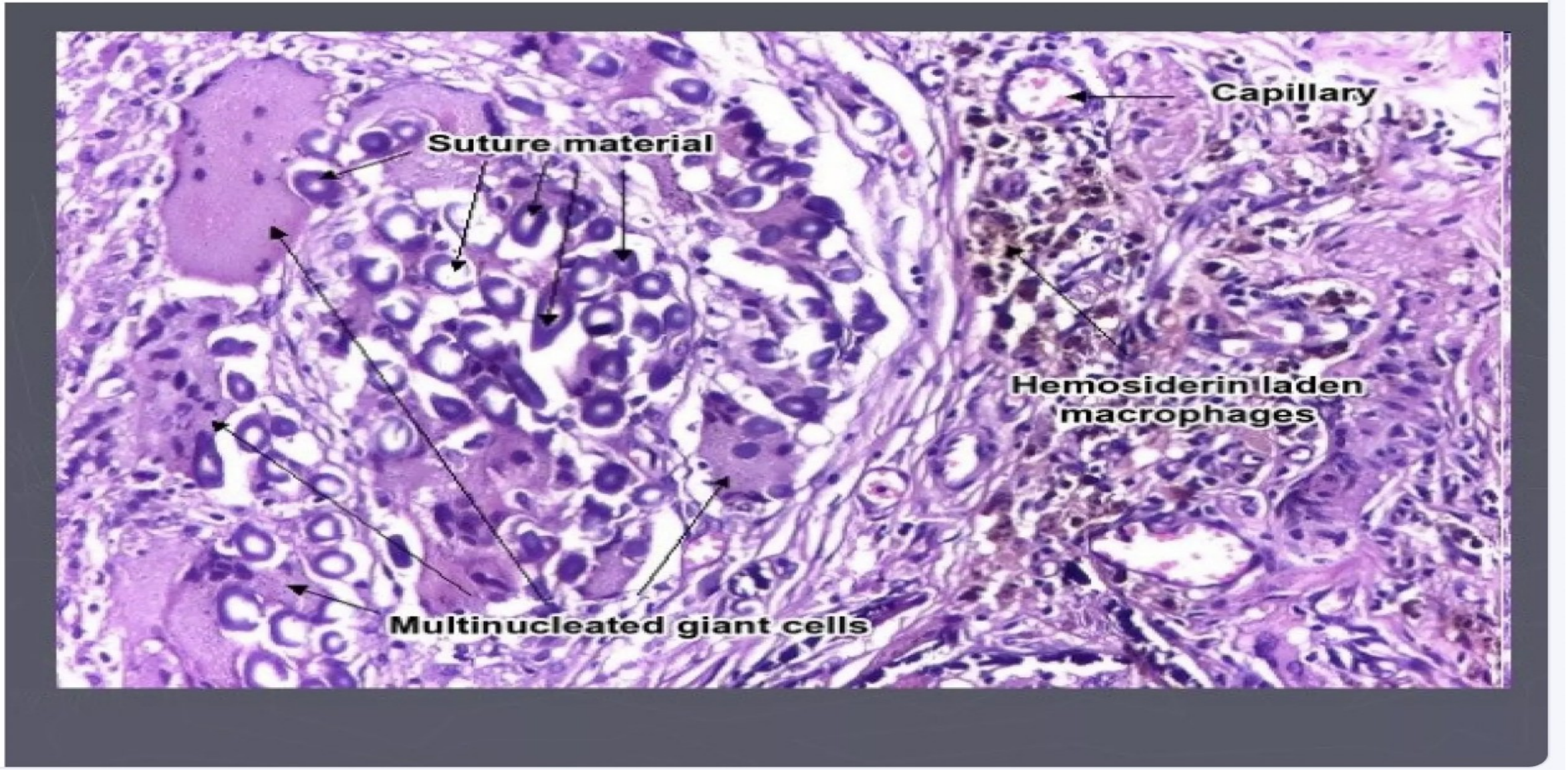
M/F of granulomatous inflammation:

- Activated macrophages in granulomas may have abundant cytoplasm resembling epithelial cells (**epithelioid histiocytes**) and fuse (**multinucleated giant cells**).
- The macrophage aggregates may be surrounded by a collar of lymphocytes & fibroblasts.
  - **CASEATING granuloma =(TB)**
    - tend to have necrotic centers
  - Mycobacterium tuberculosis often contain a central zone of necrosis +(Langhan's giant cell).
  - Grossly, this has a granular, cheesy appearance & therefore called caseous necrosis**.**
  - **NON-CASEATING granuloma**
    - Not have necrotic centers
- granulomas in Crohn disease, sarcoidosis, & foreign body reactions

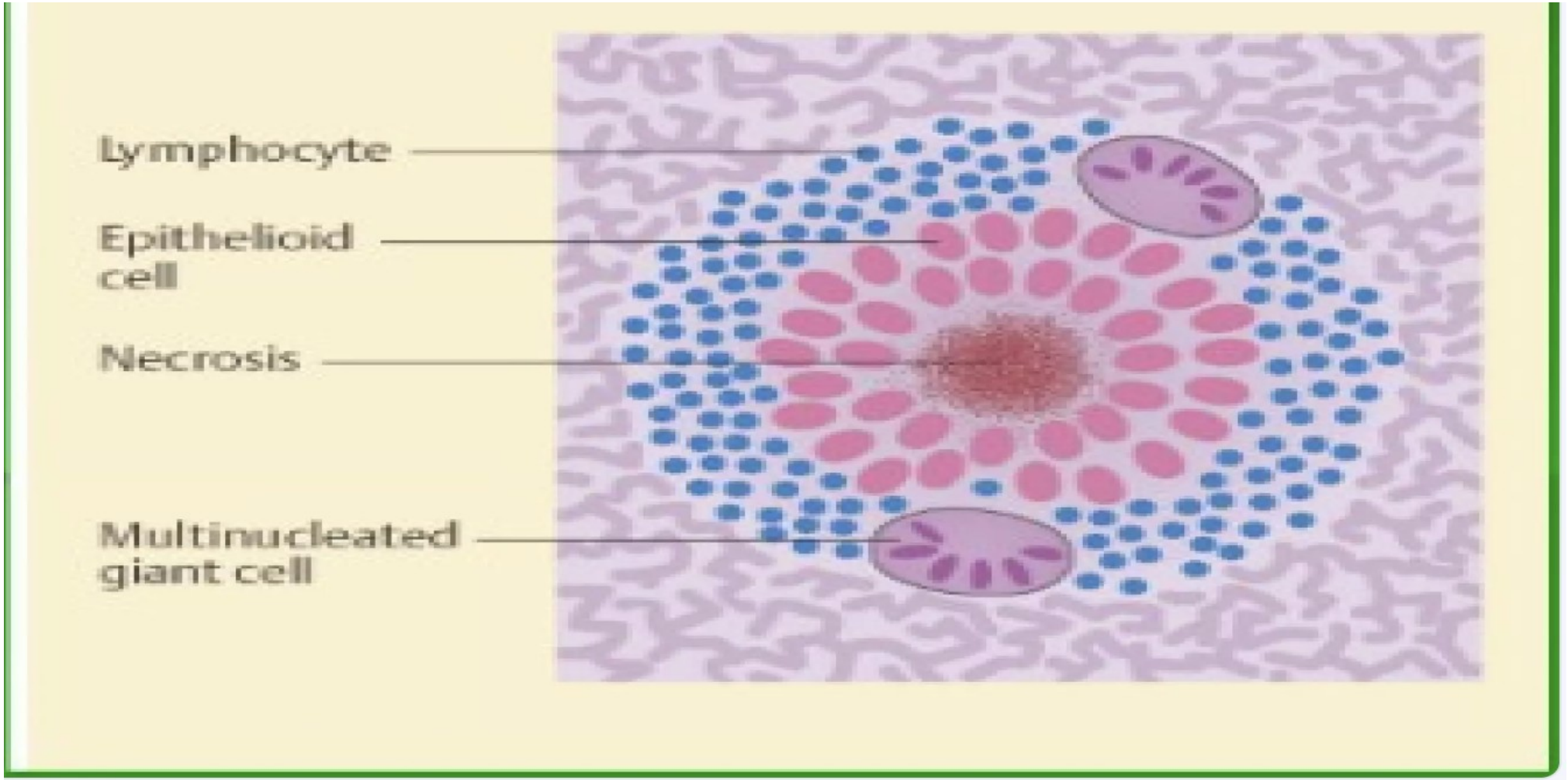
M/F of giant cells in granulomatous inflammation



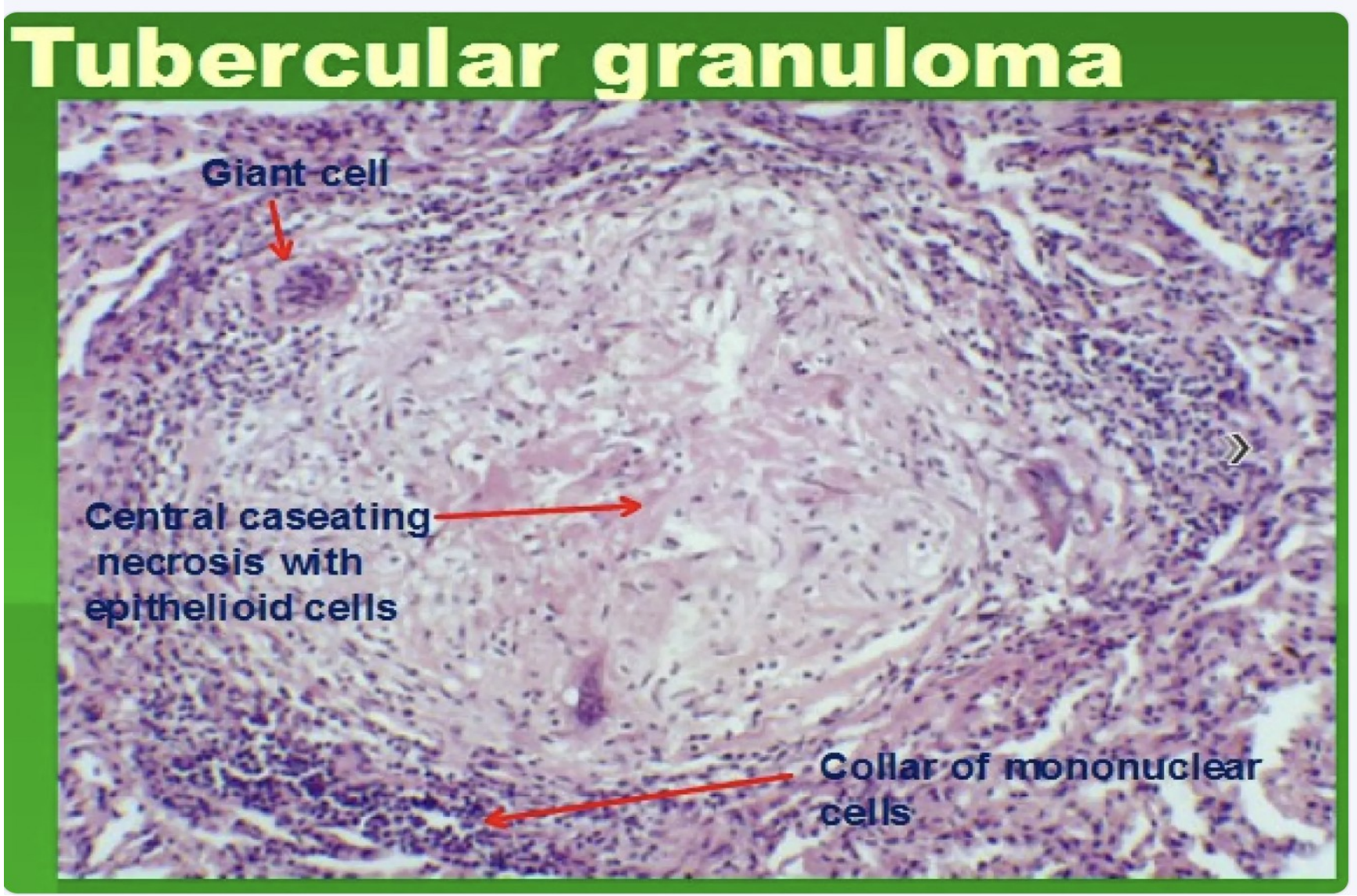
M/F of forgein body granuloma

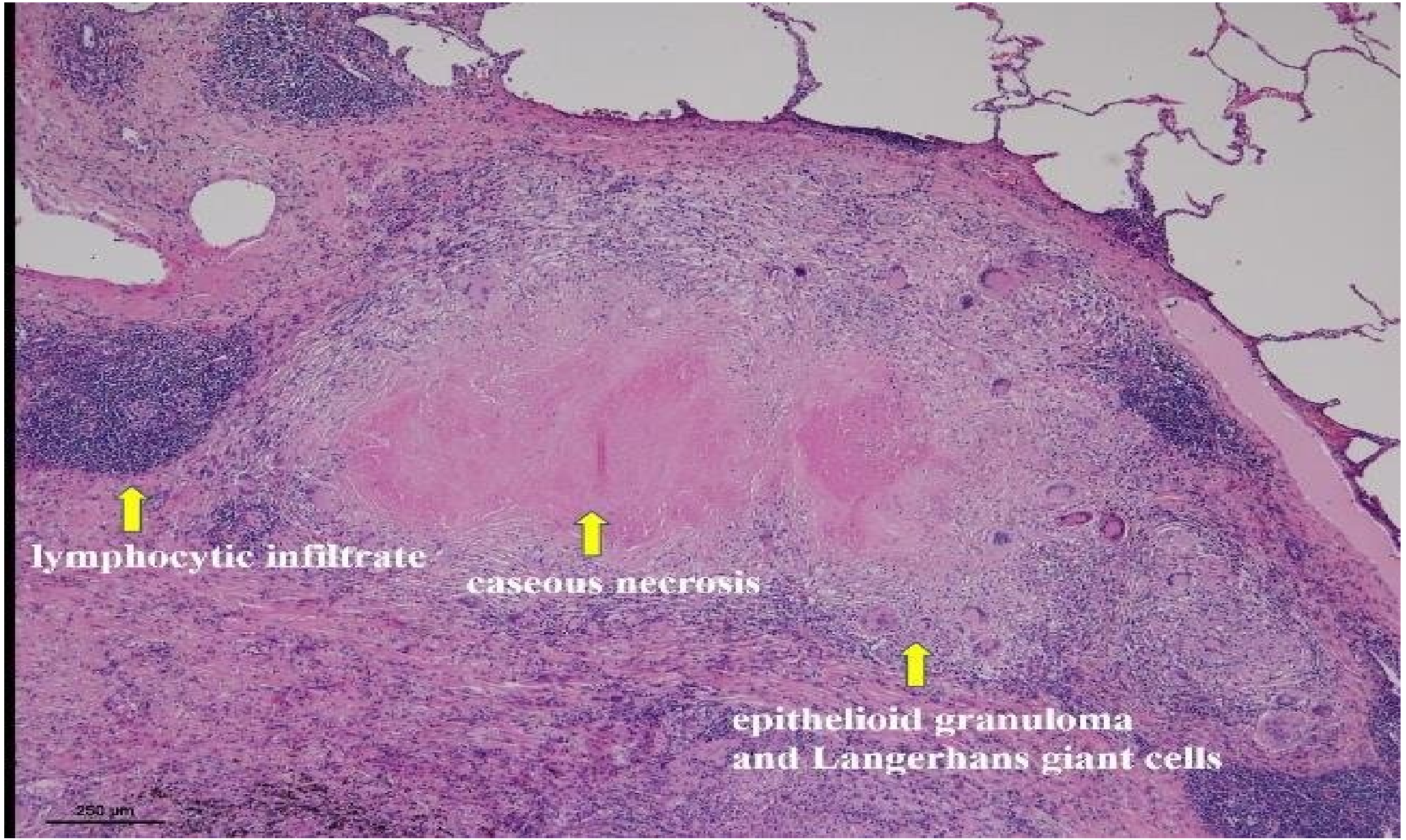


Tubercular granuloma



M/F of caseating granuloma




lymphocytic infiltrate
caseous necrosis
epithelioid granuloma
and Langerhans giant cells
250 µm

Thank You